Critical Boundaries of Quadratic Polylogarithm Feedback

Exact boundary transfer, convergent Hahn inverses, Gamma cancellations, and the marginal Lambert law

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with OpenAI assistance

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Abstract

We resolve the fixed-parameter boundary problem explicitly left open in the incoming article *Resonant Polylogarithm Transseries*. For the analytic germ defined by $uv - v^2/2 = \mathrm{Li}_s(re^v)$, we prove its exact convergence radius and unique dominant singularity at and beyond $u = \sqrt{2\zeta(s) + \zeta(s-1)^2}$, for every real $s > 2$. Beyond criticality we prove the conjectured $R^{-n}n^{-s}$ coefficient law, including its constant and every positive integer order. At criticality we construct convergent two-generator inverse charts and obtain the fractional, logarithmic, and finite-curvature regimes.

The logarithmic endpoint $s = 3$ has an explicitly normalized Lambert core. A finite coefficient formula, followed by reciprocal-Gamma differentiation, gives every inverse-logarithmic correction to its coefficient asymptotics; the first two corrections are displayed in closed form. A polylogarithmic transfer dictionary eliminates spurious descendants of logarithmic-coordinate powers. At $s = 5/2$ this proves an exact missing n^{-2} term. At $s = 4$ it instead produces a nonzero n^{-2} correction and proves that the leading square root is not a genuine Puiseux germ. We classify finite ramification at all noninteger critical orders and extend the boundary constants and marginal normalization to finite changes of the coefficient prefix.

All theorems have conventional proofs. Exact finite checks, positive coefficient recurrences, numerical diagnostics, an integration crosswalk, and further research questions accompany the article. Classical inversion, singularity analysis, and simply generated tree mechanisms are credited; worldwide priority for the model-specific refinements is not asserted.

Status. Unrefereed research manuscript, not a proof-assistant formalization. Numerical evidence is not an interval certificate. The repository observation and limits of the source audit are recorded in Section 11; no remote repository file was changed.

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1 The research question and what is added

The starting point is the quadratic feedback equation

$$uv(r) - \frac{v(r)^2}{2} = \mathrm{Li}_s\left(re^{v(r)}\right), \qquad v(0) = 0, \qquad v(r) = \sum_{n\geq 1} a_n r^n. \tag{1.1}$$

The parameters s and u are real. Our main boundary theorems assume $s > 2$ and $u \geq U_c(s)$, where

$$A = \zeta(s), \qquad B = \zeta(s-1), \qquad U_c(s) = \sqrt{2A + B^2}. \tag{1.2}$$

The variable r is complex near zero. Substituting $r = e^{-x}$ converts its Taylor series into a convergent exponential-sector expansion. Near its first singularity a different inversion problem appears, with fractional powers or logarithms of the distance to the boundary. These two levels are why the model belongs simultaneously to polylogarithm analysis and to the repository's transseries inversion program.

Equation (1.1) is a *quadratic feedback model*. Except at the dilogarithm specialization discussed in the predecessor, it is not the exact-core inversion equation for $-\mathrm{Li}_s(-e^x)$. The distinction is important when comparing this article with general integer-order Fermi inversion.

1.1 A precise pending question

The incoming article [1] proves the interior-fold regime. Its Section 12, Question 1 explicitly leaves the case $s > 2$, $u \geq U_c(s)$ open. For $u > U_c(s)$ it proposes

$$a_n \sim \frac{w^{s-1}}{(w-B)^s} R^{-n} n^{-s}, \qquad w = \sqrt{u^2 - 2A}, \quad v_b = u - w, \quad R = e^{-v_b}. \tag{1.3}$$

It correctly notes that formal transport of a singular term does not prove this assertion: one must establish complex continuation, treat resonant integer orders, and exclude other dominant singularities. We supply these steps and prove (1.3). We also determine the critical cases anticipated there.

Theorem 1.1 (Fixed-parameter boundary classification)**.** *Let $s > 2$ and $u \geq U_c(s)$ in (1.1). With w, v_b, R as in (1.3), the coefficients are strictly positive, the exact radius is R, and $r = R$ is the only singularity on $|r| = R$. The series is absolutely convergent on that circle and $v(R) = v_b$.*

For $u > U_c(s)$, (1.3) holds. For $u = U_c(s)$, so that $w = B$, the following laws hold as $n \to \infty$:

$$a_n R^n \sim \frac{(B/\Gamma(1-s))^{1/(s-1)}}{-\Gamma(-1/(s-1))} n^{-1-1/(s-1)}, \qquad 2 < s < 3, \tag{1.4}$$

$$a_n R^n \sim \sqrt{\frac{B}{\pi}}\, n^{-3/2} (\log n)^{-1/2}, \qquad s = 3, \tag{1.5}$$

$$a_n R^n \sim \sqrt{\frac{B}{2\pi(1+\zeta(s-2))}}\, n^{-3/2}, \qquad s > 3. \tag{1.6}$$

These are fixed-parameter statements, not uniform transition laws as s or u varies.

The proof is assembled in Sections 3–8. The constants in (1.4) are positive: for $1 < \alpha = s - 1 < 2$, $\Gamma(-\alpha) > 0$ and $\Gamma(-1/\alpha) < 0$.

1.2 Antecedents and proposed contribution

Polylogarithm continuation and its Gamma–zeta expansion are classical [2]. So are Lagrange inversion, the positive implicit-function schema, Hankel transfer, and logarithmic singularity analysis [5, 6]. The substitution used below places the problem in the simply generated tree/compound-Poisson setting surveyed by Janson [7]. The repository already has a linear-core critical Hahn theory, with ramification and arithmetic cancellation results, and related marginal and confluent packages [8, 9]. Those mechanisms are not discoveries of this article.

The proposed additions are narrower and explicit: completion of the predecessor's quadratic boundary theorem; its all-order two-variable charts on both sides of $s = 3$; the exact marginal Lambert normalization and finite Gamma-derivative coefficient compiler; the $s = 5/2$ and $s = 4$ coefficient identities; and a finite-prefix dictionary that tracks the quadratic correction instead of importing linear-core constants unchanged. The explicit square-root lift below explains both the connection with classical theory and the extra constants it introduces.

2 The positive quadratic lift and exact coefficients

Define, with the square root positive at $t = 0$,

$$h(t) = u - \sqrt{u^2 - 2\,\mathrm{Li}_s(t)}, \qquad t = re^{h(t)}, \qquad v(r) = h(t(r)). \tag{2.1}$$

Since $h(0) = 0$ and $u > 0$, the analytic implicit function theorem gives a unique germ $t(r)$, and hence $v(r)$, near zero.

Proposition 2.1 (Three exact coefficient descriptions)**.** *For any real s and $u > 0$, let $h(t) = \sum_{j\ge1} h_j t^j$ be the lift germ. Then*

$$uh_j = j^{-s} + \frac{1}{2}\sum_{k=1}^{j-1} h_k h_{j-k}, \tag{2.2}$$

$$a_n = \frac{1}{n}[t^n]\exp(nh(t)), \tag{2.3}$$

$$a_n = \sum_{k=1}^{n} \frac{C_{n,k}^{(s)}}{k} \sum_{j=0}^{k-1} \binom{k+j-1}{j} \frac{n^{k-1-j}}{(k-1-j)!\,2^j u^{k+j}}, \tag{2.4}$$

$$C_{n,k}^{(s)} = [z^n]\,\mathrm{Li}_s(z)^k = \sum_{\substack{n_1+\cdots+n_k=n \\ n_j\ge1}} (n_1\cdots n_k)^{-s}.$$

In particular $h_j > 0$ and $a_n > 0$. Formula (2.4) is the predecessor's positive finite formula, restated for comparison rather than claimed anew.

Proof. The identity $uh - h^2/2 = \mathrm{Li}_s(t)$ gives (2.2), and induction proves positivity. Lagrange inversion in $t = re^{h(t)}$ gives

$$[r^n]h(t(r)) = \frac{1}{n}[t^{n-1}]h'(t)e^{nh(t)} = \frac{1}{n}[t^n]e^{nh(t)}.$$

For the finite formula, rewrite (1.1) as $v = \mathrm{Li}_s(re^v)/(u - v/2)$, insert an auxiliary multiplier, and apply Lagrange inversion in that multiplier. The coefficient of r^n in the kth power of the numerator is $C_{n,k}^{(s)}e^{nv}$ and vanishes for $k > n$. Expanding $(u - v/2)^{-k}$ and the exponential gives (2.4). Positivity also follows directly from either of the last two expressions. □

Corollary 2.2 (An exact positive coefficient identity)**.** *If $s > 2$ and $u \ge U_c(s)$, then*

$$nR^n a_n = [t^n]\exp\big(n(h(t) - h(1))\big). \tag{2.5}$$

The right side is the probability that $\sum_{j\ge1} jN_j = n$, where the independent N_j have Poisson means nh_j. Their total mean count is $nh(1) < \infty$.

Proof. On $|t| \le 1$, $2|\operatorname{Li}_s(t)|/u^2 \le 2A/u^2 < 1$. Thus the lift is well defined and $h(1) = u - w = v_\mathrm{b}$. Multiply (2.3) by $n\mathrm{e}^{-nv_\mathrm{b}}$. The exponential of $n\sum h_j(t^j - 1)$ is the stated probability generating function. □

This identity provides a positive numerical recurrence without subtracting large alternating terms. It is also an exact bridge to classical random allocation methods; no probability limit theorem is needed for the proofs of the fixed-parameter laws below.

3 Global control up to the boundary

Proposition 3.1 (No hidden singularity inside the proposed circle)**.** *Suppose $s > 2$ and $u \ge U_\mathrm{c}(s)$. Then the germ v extends analytically to $|r| < R$, its Taylor series converges absolutely on $|r| \le R$, and*

$$\sum_{n\ge1} a_n R^n = v_\mathrm{b}. \tag{3.1}$$

It extends analytically through every point of $|r| = R$ except possibly $r = R$.

Proof. For $0 \le t \le 1$, put

$$v(t) = u - \sqrt{u^2 - 2\operatorname{Li}_s(t)}, \qquad r(t) = t\mathrm{e}^{-v(t)}.$$

The square root is at least $w \ge B > 0$. On $0 < t < 1$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\log r(t) = \frac{1}{t}\left(1 - \frac{\operatorname{Li}_{s-1}(t)}{\sqrt{u^2 - 2\operatorname{Li}_s(t)}}\right) > 0. \tag{3.2}$$

Indeed $\operatorname{Li}_{s-1}(t) < B \le w$. Hence this parameterization supplies the real solution on $0 \le r < R$, tending to v_b as $r \uparrow R$.

To pass from a real branch to its complex Taylor disk, iterate

$$v_0 = 0, \qquad v_{j+1} = \Phi(r, v_j), \qquad \Phi(r, v) = \frac{v^2}{2u} + \frac{\operatorname{Li}_s(r\mathrm{e}^v)}{u}.$$

All coefficients are nonnegative and increase coefficientwise. Each fixed coefficient stabilizes: if two successive iterates agree modulo r^k, their images agree modulo r^{k+1}, since $\Phi_v(0,0) = 0$ and both iterates vanish at zero. For each fixed $0 \le r < R$, the iterates are bounded by the real solution, since Φ preserves order on the interval between zero and that solution. Monotone convergence of the coefficients therefore proves convergence of the full formal series at every such r. Its sum agrees with the real solution near zero, and then on the full interval by analytic continuation. Taking $r \uparrow R$ proves (3.1). Nonnegative coefficients imply absolute and uniform convergence on the closed disk.

Let $r_0 \ne R$ lie on $|r| = R$. Since every a_n is positive, strict triangle inequality gives $|v(r_0)| < v_\mathrm{b}$. To justify strictness, equality would force all r_0^n/R^n, $n \ge 1$, to have the same argument, which the terms $n = 1, 2$ already exclude unless $r_0 = R$. Consequently

$$|r_0\mathrm{e}^{v(r_0)}| = R\mathrm{e}^{\Re v(r_0)} < R\mathrm{e}^{v_\mathrm{b}} = 1.$$

The equation is analytic there, and

$$\begin{aligned}|\Phi_v(r_0, v(r_0))| &\le \frac{|v(r_0)| + \operatorname{Li}_{s-1}(|r_0\mathrm{e}^{v(r_0)}|)}{u}\\ &< \frac{v_\mathrm{b} + B}{u} \le 1.\end{aligned}$$

The analytic implicit function theorem gives continuation through r_0. This proves the proposition without inferring a complex radius from the real parameterization alone. □

The local singular expansions proved below are valid in a sector at R with opening larger than π. Together with the proposition, compactness of the remaining circle supplies the usual dented-disk, or Δ-domain, continuation. A nonzero local singular term then makes R the exact radius and the unique dominant singularity. This is the analytic justification required before applying coefficient transfer.

4 Logarithmic coordinates and a transfer dictionary

Near the boundary use the determinations real on $0 < r < R$:

$$X = -\log(r/R), \qquad \tau = -\log t, \qquad v = v_{\mathrm{b}} + X - \tau. \tag{4.1}$$

The exact inverse relation is

$$X = H(\tau) := \tau - \left(\sqrt{u^2 - 2\,\mathrm{Li}_s(\mathrm{e}^{-\tau})} - w\right). \tag{4.2}$$

For $s \notin \mathbb{N}$, the classical convergent expansion is

$$\mathrm{Li}_s(\mathrm{e}^{-\tau}) = \Gamma(1-s)\tau^{s-1} + \sum_{k\geq 0} \frac{(-1)^k \zeta(s-k)}{k!}\tau^k, \qquad |\tau| < 2\pi, \quad |\arg\tau| < \pi. \tag{4.3}$$

Its source is DLMF 25.12.12 [2]. At a positive integer m, the cancelling Gamma and zeta poles give

$$\mathrm{Li}_m(\mathrm{e}^{-\tau}) = \frac{(-\tau)^{m-1}}{(m-1)!}(H_{m-1} - \log\tau) + \sum_{\substack{k\geq 0 \\ k\neq m-1}} \frac{(-1)^k \zeta(m-k)}{k!}\tau^k, \tag{4.4}$$

where $H_j = \sum_{k=1}^{j} 1/k$. This follows either by taking the cancelling limit in (4.3), or by differentiating in τ and matching the elementary $m = 1$ identity.

Expanding the square in (1.1) after (4.1), rather than prematurely expanding an inverse, gives the exact local equation, for noninteger s,

$$\begin{aligned} wX = (w-B)\tau + \Gamma(1-s)\tau^{s-1} + \frac{1+\zeta(s-2)}{2}\tau^2 - \tau X + \frac{X^2}{2} \\ + \sum_{k\geq 3} \frac{(-1)^k \zeta(s-k)}{k!}\tau^k. \end{aligned} \tag{4.5}$$

The additional 1 in the quadratic coefficient is the contribution of $-v^2/2$. It must not be dropped when importing a linear-core formula.

4.1 Why powers of X have no spurious descendants

Lemma 4.1 (Polylogarithmic transfer coordinates)**.** *Let a function analytic at $r = 0$ have a unique dominant point R and a Δ-continuation there. Write $X = -\log(r/R)$ locally. A singular term cX^λ, $\lambda \notin \mathbb{N}_0$, contributes*

$$\frac{c}{\Gamma(-\lambda)} R^{-n} n^{-\lambda-1} \tag{4.6}$$

to coefficient asymptotics, without an additional string of integer $1/n$ descendants from that term alone. A term cX^k, $k \in \mathbb{N}_0$, is locally analytic and contributes no algebraic singularity term. A term $cX^k \log X$ contributes

$$c(-1)^{k+1} k!\, R^{-n} n^{-k-1}. \tag{4.7}$$

More generally, for fixed j the contribution of $X^\lambda(\log X)^j$ is obtained by applying ∂_λ^j to $R^{-n}n^{-\lambda-1}/\Gamma(-\lambda)$.

These statements concern local contributions to an asymptotic expansion; X^λ need not itself be an analytic germ at $r = 0$.

Proof. Put $z = r/R$. For $\lambda \notin \mathbb{N}_0$, the convergent identity (4.3) gives

$$\mathrm{Li}_{\lambda+1}(z) = \Gamma(-\lambda)(-\log z)^\lambda + \text{a function analytic near } z = 1.$$

Subtract $c\,\mathrm{Li}_{\lambda+1}(z)/\Gamma(-\lambda)$ from the original function. The subtracted function has the exact nth coefficient

$$cn^{-\lambda-1}/\Gamma(-\lambda).$$

The indicated singular term has therefore been removed. A finite collection of such subtractions reduces the remainder to the ordinary big-O transfer theorem [5, 6]; any local analytic part can first be removed to the required order by a polynomial in $1 - z$. This proves the absence of artificial descendants.

For integer k, (4.4) gives the same conclusion using Li_{k+1}. Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\frac{1}{\Gamma(-\lambda)}\bigg|_{\lambda=k} = (-1)^{k+1}k!.$$

For logarithmic powers, differentiate the normally convergent local identity in a parameter neighborhood not containing a pole, or use its pole-cancelled integer limit. Local analytic terms remain irrelevant to any fixed algebraic asymptotic order. This proves the finite-order statement, not convergence of an infinite coefficient-asymptotic series. □

Remark 4.2 (Cancellation versus a different coordinate). Expanding X^λ first in powers of $1 - r/R$ produces many terms. Applying ordinary power transfer to those terms separately is legitimate only after combining all terms of the desired order. The apparent integer $1/n$ corrections cancel. Lemma 4.1 performs this cancellation exactly by using a polylogarithm as the transfer model.

4.2 Local inverse sectors

All subsequent local estimates are uniform on each closed subsector $|\arg X| \le \pi - \epsilon$, $0 < |X| < \delta_\epsilon$. For a simple leading relation $X = c\tau^p(1 + o(1))$, with $c > 0$ and $p \ge 1$, the selected inverse is obtained near $\tau = (X/c)^{1/p}$, real positive for $X > 0$. One may prove sectorial inversion by Rouché's theorem on a small relative circle about this root: the leading map has one simple root there, and the perturbation and its first derivative are smaller uniformly. The root depends holomorphically on X away from zero, and uniqueness glues the local choices. The explicit implicit-function charts below give a stronger convergent construction when s is noninteger. The logarithmic case is treated separately with its Lambert core.

5 The strict boundary theorem, including integer orders

Theorem 5.1 (Proof of the pending boundary law)**.** *Let $s > 2$ and $u > U_c(s)$. Set $\lambda = 1 - B/w > 0$. Then the exact radius and unique dominant point are $R = \mathrm{e}^{-(u-w)}$, and*

$$a_n = \frac{1}{w\lambda^s}R^{-n}n^{-s}(1 + o(1)). \tag{5.1}$$

For noninteger s, the leading nonanalytic part of v in X coordinates is

$$\frac{\Gamma(1-s)}{w\lambda^s}X^{s-1}. \tag{5.2}$$

For integer $m \geq 3$, it is

$$\frac{(-1)^m}{(m-1)!\, w\lambda^m} X^{m-1} \log X. \tag{5.3}$$

In each case the omitted local terms are analytic or have strictly higher singular order.

Proof. First suppose s is noninteger. Remove the fractional term from (4.3), leaving an analytic function $Z(\tau)$, and let

$$H_0(\tau) = \tau - \left(\sqrt{u^2 - 2Z(\tau)} - w\right).$$

Then $H_0(0) = 0$, $H_0'(0) = \lambda$, and its ordinary analytic inverse $\tau_0(X)$ satisfies $\tau_0(X) = X/\lambda + O(X^2)$. Expanding the square root about its nonzero value w gives

$$H(\tau) = H_0(\tau) + \frac{\Gamma(1-s)}{w}\tau^{s-1} + O(\tau^s) + O(\tau^{2s-2})$$

on the chosen sectors. Sectorial inversion, or substitution followed by Rouché's theorem, therefore gives

$$\tau(X) = \tau_0(X) - \frac{\Gamma(1-s)}{w\lambda^s} X^{s-1} + O(X^{s-1+\eta})$$

for any fixed $0 < \eta < \min(1, s-2)$. This bound permits analytic terms to be absorbed in τ_0; the nonlinear interaction of two leading fractional corrections first has order $2s - 3 > s - 1$. The identity $v = v_{\mathrm{b}} + X - \tau$ proves (5.2).

At an integer $m \geq 3$, use (4.4) instead. Its analytic part again has a simple inverse, and its leading nonanalytic term is $\ell_m \tau^{m-1} \log \tau$, where $\ell_m = (-1)^m/(m-1)!$. The same calculation gives (5.3); terms involving $\log \lambda$ are analytic at this order. The remaining nonanalytic terms are of higher order, with at most fixed logarithmic factors, so they are negligible in the leading coefficient asymptotic.

The nonzero singular terms show that $r = R$ is singular. Proposition 3.1 excludes a smaller radius and any other singularity on that circle. The sectorial inverse and the compact continuation away from R supply a Δ-domain. Lemma 4.1 now gives (5.1) in both cases. Finally $1/(w\lambda^s) = w^{s-1}/(w-B)^s$, which is precisely (1.3). □

Remark 5.2 (The singular limit in the theorem). The constant in (5.1) diverges as $w \downarrow B$. Thus substituting $u = U_{\mathrm{c}}(s)$ in this formula is not valid. Criticality changes the inverse equation's leading order, not merely its prefactor.

6 The fractional critical interval: a convergent Hahn chart

Assume $u = U_{\mathrm{c}}(s)$ and $2 < s < 3$. Write

$$\alpha = s - 1 \in (1, 2), \qquad G = \Gamma(-\alpha) > 0, \qquad d = (B/G)^{1/\alpha}, \qquad T = dX^{1/\alpha}. \tag{6.1}$$

Set $\rho = T^{\alpha-1}$ and $\sigma = T^{2-\alpha}$; thus $T = \rho\sigma$. Both generators tend to zero.

Theorem 6.1 (Convergent two-generator critical inverse)**.** *There is a unique analytic function $U(\rho, \sigma)$ near $(0, 0)$ with $U(0, 0) = 1$ such that the boundary inverse is*

$$\tau = TU(\rho, \sigma), \qquad v = v_{\mathrm{b}} + X - TU(\rho, \sigma). \tag{6.2}$$

It is determined by the analytic equation

$$\begin{aligned} 1 = U^\alpha + \frac{1 + \zeta(\alpha - 1)}{2G}\sigma U^2 - \frac{\rho\sigma}{B}U + \frac{G}{2B^2}\rho^2\sigma \\ + \frac{\sigma}{G}\sum_{k \geq 3} \frac{(-1)^k \zeta(\alpha + 1 - k)}{k!}(\rho\sigma)^{k-2}U^k. \end{aligned} \tag{6.3}$$

Its Taylor series gives an absolutely convergent generalized power series for τ, with support contained in

$$\left\{\frac{1+j(\alpha-1)+k(2-\alpha)}{\alpha} : j,k\in\mathbb{N}_0\right\}. \tag{6.4}$$

Only finitely many support exponents lie below any fixed bound; collisions are combined by addition.

If $\alpha = p/q$ in lowest terms, the exact minimum ramification index is p: the inverse is a convergent Puiseux series in $X^{1/p}$, and cannot be expressed as a convergent power series in $X^{1/M}$ with a smaller positive integer M. If α is irrational, no finite Puiseux ramification is possible.

Proof. At criticality $w = B$. In (4.5) put $X = (G/B)T^\alpha$ and $\tau = TU$, and divide by GT^α. The identities $T^{2-\alpha} = \sigma$, $T = \rho\sigma$, and $T^\alpha = \rho^2\sigma$ give exactly (6.3). The tail is analytic near $(\rho,\sigma,U) = (0,0,1)$ because (4.3) has a convergent analytic tail. Choose the power U^α analytic near $U = 1$. The derivative of the right side with respect to U is $\alpha \neq 0$ at the base point. The analytic implicit function theorem therefore produces U and an absolutely convergent bivariate Taylor series on a polydisk.

Substitution of the two small generators gives absolute convergence on closed subsectors and the support (6.4). Its nonnegative integer generators have positive weights, proving local finiteness. The branch agrees with the real inverse for $X > 0$ small, by its leading behavior, and hence with the continued germ selected in Section 3.

For $\alpha = p/q$, the exponents are $\{(q + j(p-q) + k(2q-p))/p\}$. They are nonnegative integer multiples of $1/p$, and absolute convergence gives an analytic series in $X^{1/p}$. The leading term $dX^{q/p}$ is nonzero, so any possible ramification index must be divisible by p. For irrational α, its leading exponent $1/\alpha$ is irrational and already excludes finite ramification. □

6.1 A finite formula for every chart coefficient

The preceding implicit function is not merely existential. The following finite recipe applies to both critical charts in this paper.

Lemma 6.2 (Finite bivariate reversion)**.** *Suppose $E(p,q,U) = E_0(U) + \Delta E(p,q,U)$ is analytic near $(0,0,1)$, $E_0(1) = 0$, $E_0'(1) \neq 0$, and $\Delta E(0,0,U) = 0$. Let $U(0,0) = 1$ solve $E = 0$, and define*

$$Q(z) = \frac{E_0(1+z)}{z}, \qquad \Delta(p,q,z) = \Delta E(p,q,1+z).$$

For $j + k \geq 1$,

$$[p^j q^k](U-1) = \sum_{\ell=1}^{j+k} \frac{(-1)^\ell}{\ell}[p^j q^k z^{\ell-1}]\left(\frac{\Delta(p,q,z)}{Q(z)}\right)^\ell. \tag{6.5}$$

Thus each coefficient requires only finite Taylor data.

Proof. Writing $z = U - 1$ converts the equation to $z = -\Delta(p,q,z)/Q(z)$. Insert a multiplier a on the right and apply the ordinary Lagrange formula in a. Its ℓth coefficient is $(-1)^\ell[z^{\ell-1}](\Delta/Q)^\ell/\ell$. Since Δ has positive total (p,q) degree, terms with $\ell > j + k$ cannot contribute to $p^j q^k$. Setting $a = 1$ is therefore legitimate coefficientwise, giving the finite formula. Analyticity follows independently from the implicit function theorem. □

For (6.3), take $E_0(U) = U^\alpha - 1$ and let ΔE be its displayed parameter-dependent part. This makes every Hahn coefficient a finite expression in Gamma values, zeta values, and rational functions of α and B.

Corollary 6.3 (Coefficient expansion to every finite order)**.** *Combine the collisions in Theorem 6.1 and write $\tau = \sum_\lambda c_\lambda X^\lambda$. For any positive L outside the grid and outside $\mathbb{N}_0$,*

$$a_n R^n = -\sum_{\lambda<L} \frac{c_\lambda}{\Gamma(-\lambda)} n^{-\lambda-1} + O(n^{-L-1}). \tag{6.6}$$

The reciprocal Gamma factor is zero at nonnegative integer λ. In particular (1.4) holds, and R is the exact radius and unique dominant point.

Proof. Absolute convergence of the two-variable chart and positivity of the two weights imply that truncating below L leaves $O(X^L)$ on each closed subsector. One way to see the tail bound is to fix a smaller polydisk, apply its geometric Taylor majorant, and factor $|X|^L$ from all omitted monomials. Local finiteness leaves a positive gap before the first omitted exponent. The leading term $dX^{1/\alpha}$ is nonanalytic and nonzero. Proposition 3.1 and the chart supply exact dominance and Δ-continuation. Apply Lemma 4.1 to the finite truncation and then big-O transfer to its remainder. The analytic term $v_\mathrm{b} + X$ contributes nothing at these algebraic orders. □

The expansion in (6.2) is genuinely convergent. Equation (6.6), in contrast, is asserted to every *fixed finite order*; no convergence claim is made for its infinite coefficient-asymptotic sum.

6.2 An exact cancellation at order $s = 5/2$

Corollary 6.4 (The missing n^{-2} term)**.** *At $s = 5/2$, $u = U_\mathrm{c}(5/2)$, put*

$$B = \zeta(3/2), \quad G = \Gamma(-3/2) = \frac{4\sqrt{\pi}}{3}, \quad d = (B/G)^{2/3}, \quad c = \frac{1+\zeta(1/2)}{2G}, \quad K_* = \frac{7c^2}{9} + \frac{2}{3B}. \tag{6.7}$$

Then

$$a_n R^n = -\frac{d}{\Gamma(-2/3)} n^{-5/3} - \frac{K_* d^2}{\Gamma(-4/3)} n^{-7/3} + O(n^{-8/3}). \tag{6.8}$$

The coefficient of n^{-2} is exactly zero. The second displayed coefficient is strictly negative.

Proof. Here $\rho = \sigma = T^{1/2}$. Writing $z = T^{1/2}$ in (6.3), its first two coefficient equations are

$$1 = U^{3/2} + czU^2 - z^2 U/B + O(z^3).$$

Solving them gives

$$U = 1 - \frac{2c}{3} T^{1/2} + \left(\frac{7c^2}{9} + \frac{2}{3B}\right) T + O(T^{3/2}). \tag{6.9}$$

Thus

$$v = v_\mathrm{b} + X - T + \frac{2c}{3} T^{3/2} - K_* T^2 + O(T^{5/2}).$$

Since $T = dX^{2/3}$, the term $T^{3/2} = d^{3/2}X$ is analytic. It is annihilated by $1/\Gamma(-1) = 0$ in the transfer dictionary, rather than producing a coefficient term of order n^{-2}. The other terms give (6.8). Finally $K_* > 0$ and $\Gamma(-4/3) > 0$. □

This is an arithmetic cancellation in a specified quadratic model, not a new general discovery that analytic local terms vanish under singularity transfer. The related mechanism is already present in the repository's linear-core critical Hahn theory [8].

7 The marginal order three and its complete logarithmic hierarchy

Now let $s = 3$, $u = U_c(3)$, and $B = \zeta(2)$. Pole cancellation in (4.4) gives

$$\mathrm{Li}_3(\mathrm{e}^{-\tau}) = \zeta(3) - B\tau + \frac{\tau^2}{2}\left(\frac{3}{2} - \log\tau\right) + \frac{\tau^3}{12} + O(\tau^4). \tag{7.1}$$

The quadratic equation therefore has

$$X = \frac{\tau^2}{2B}\left(\log\frac{1}{\tau} + \frac{5}{2}\right) + O\left(\tau^3 \log\frac{1}{\tau}\right). \tag{7.2}$$

The constant is $5/2$, not $3/2$. The extra 1 is the quadratic-core contribution in (4.5).

Theorem 7.1 (The exact Lambert core)**.** *Define the large positive solution Y by*

$$Y - \log Y = \log(1/X) + 5 - \log(4B). \tag{7.3}$$

For positive small X this is equivalently

$$Y = -W_{-1}(-4B\mathrm{e}^{-5}X), \qquad \tau_0(X) = 2\sqrt{\frac{BX}{Y}}. \tag{7.4}$$

Then the true inverse satisfies, on the selected complex sectors,

$$\tau(X) = \tau_0(X) + O(\tau_0(X)^2). \tag{7.5}$$

The large solution of (7.3), continued from $X > 0$, defines the complex branch in this statement; a numerical Lambert branch index is not to be used without that continuation convention.

Proof. Let $H_0(\tau) = \tau^2(\log(1/\tau) + 5/2)/(2B)$. Setting $Y = 2\log(1/\tau_0) + 5$ converts $H_0(\tau_0) = X$ into $Y\mathrm{e}^{-Y} = 4B\mathrm{e}^{-5}X$, proving (7.4). The large solution is holomorphic for $\Re\log(1/X)$ sufficiently large and bounded imaginary part: the derivative of $Y - \log Y$ is $1 - 1/Y$, uniformly separated from zero. Its real determination is the usual W_{-1} branch [3].

Uniformly on closed subsectors, $|H_0'(\tau_0)| \asymp |\tau_0 \log\tau_0|$ and $H(\tau) - H_0(\tau) = O(\tau^3\log\tau)$ by (7.2). On a circle $|\tau - \tau_0| = C|\tau_0|^2$, the linear part of $H_0(\tau) - H_0(\tau_0)$ dominates this perturbation when C is chosen large and then X small. Higher Taylor terms are smaller. Rouché's theorem gives one root there; its derivative is nonzero. The roots glue holomorphically and match the physical real inverse. This proves (7.5) and provides the required sectorial continuation. □

The error in (7.5) is power-smaller than the leading square root. It therefore lies beyond every fixed inverse-logarithmic order in the coefficient expansion. This is stronger than retaining just the leading regularly varying factor.

7.1 All coefficients by finite Lagrange and Gamma formulas

Let $I(a) = 1/\Gamma(-a)$ and define

$$g_j = \frac{I^{(j)}(1/2)}{I(1/2)}, \qquad g_0 = 1. \tag{7.6}$$

For example,

$$g_1 = p := \psi(-1/2) = 2 - \gamma - 2\log 2, \qquad g_2 = p^2 - \psi^{(1)}(-1/2), \qquad \psi^{(1)}(-1/2) = 4 + \frac{\pi^2}{2}. \tag{7.7}$$

These are exact Gamma and polygamma constants, not fitted parameters. Define polynomials Q_k and P_m by

$$Q_0(A) = 1,$$
$$Q_k(A) = -\frac{1}{2k}[z^{k-1}](1+z)^{-3/2}(A+\log(1+z))^k, \qquad k \geq 1, \tag{7.8}$$
$$P_m(A) = \sum_{j=0}^{m} \frac{(-1)^j g_j}{j!} \left\{ \prod_{\ell=0}^{j-1} \left(\frac{\mathrm{d}}{\mathrm{d}A} - m + j - \ell - \frac{1}{2} \right) \right\} Q_{m-j}(A). \tag{7.9}$$

The product is the identity when $j = 0$.

Theorem 7.2 (Every inverse-logarithmic coefficient)**.** *Let $L = \log n$ and $A_n = \log L + 5 - \log(4B)$. For each fixed integer $M \geq 0$,*

$$a_n R^n = \sqrt{\frac{B}{\pi}}\, n^{-3/2} L^{-1/2} \left\{ \sum_{m=0}^{M} \frac{P_m(A_n)}{L^m} + O\left(\frac{(\log L)^{M+1}}{L^{M+1}} \right) \right\}. \tag{7.10}$$

Each P_m has degree m, with leading coefficient $\binom{-1/2}{m}$. In particular, with

$$D_n = \log\log n + 3 + \gamma - \log B, \tag{7.11}$$

$$a_n R^n = \sqrt{\frac{B}{\pi}}\, n^{-3/2} L^{-1/2} \left[1 - \frac{D_n}{2L} + \frac{3D_n^2/8 - D_n/2 - \frac{3}{8}(4 + \pi^2/2)}{L^2} + O\left(\frac{(\log L)^3}{L^3} \right) \right]. \tag{7.12}$$

The leading term is nonanalytic, so R is the exact radius and unique dominant point.

Proof. Put $\mathcal{L} = \log(1/X)$, $c = 5 - \log(4B)$, $A = \log \mathcal{L} + c$, and write $Y = \mathcal{L}H$ in (7.3). Then

$$H = 1 + \frac{A + \log H}{\mathcal{L}}. \tag{7.13}$$

For fixed A, ordinary Lagrange inversion for $H - 1$ gives

$$H^{-1/2} = \sum_{k \geq 0} Q_k(A) \mathcal{L}^{-k}$$

with (7.8). Along $A = \log \mathcal{L} + c$, this expansion to any fixed order has remainder $O(\mathcal{L}^{-M-1}(1 + |A|)^{M+1})$. To justify uniformity, treat $p = 1/\mathcal{L}$ and $z = A/\mathcal{L}$ as independent variables in $H = 1 + z + p \log H$. The solution is analytic near $(p, z) = (0, 0)$; its convergent double Taylor majorant supplies the asserted remainder when both variables tend to zero. The same argument controls any fixed number of derivatives.

For completeness, the finite logarithmic transfer calculation can be made directly in X coordinates. In a Cauchy coefficient integral use $r = R\mathrm{e}^{-X}$ and then $X = z/n$. The local singular integral has the Hankel kernel $\mathrm{e}^z z^{1/2}\,\mathrm{d}z$ with no further $1/n$ Jacobian corrections. If $f(\mathcal{L})$ denotes a slowly varying factor, its finite Taylor expansion at L is

$$f(L - \log z) = \sum_{j=0}^{J} \frac{(-1)^j}{j!} f^{(j)}(L)(\log z)^j + \text{remainder}.$$

The Hankel integral of $\mathrm{e}^z z^{1/2}(\log z)^j$ divided by $2\pi\mathrm{i}$ is $I^{(j)}(1/2)$. This is the reciprocal-Gamma Hankel formula and its fixed derivatives. Thus the normalized transfer operator is

$$f(L) \longmapsto \sum_{j\ge 0} \frac{(-1)^j g_j}{j!} f^{(j)}(L), \tag{7.14}$$

used only to the finite order required.

Here $f(\mathcal{L}) = \sum_k Q_k(\log \mathcal{L} + c)\mathcal{L}^{-k-1/2}$. Differentiating one term j times gives

$$\mathcal{L}^{-k-j-1/2} \prod_{\ell=0}^{j-1} \left(\frac{\mathrm{d}}{\mathrm{d}A} - k - \ell - \frac{1}{2} \right) Q_k(A).$$

Collect $k + j = m$ in (7.14); this is precisely (7.9). Multiplication by the singular prefactor $-2\sqrt{B}$ and by $I(1/2) = -1/(2\sqrt{\pi})$ gives (7.10).

The finite Hankel expansion has the stated uniform remainder. One can split the scaled contour into $|\log z| \le L/2$ and its complement. On the first part the analytic majorants above and the Taylor remainder bound give the next inverse-logarithmic order after integration. Near $z = 0$, the factor $z^{1/2}$ makes the excluded portion exponentially small in L; at infinity the tilted Hankel rays give exponential decay in $|z|$. The part of the original contour away from R is exponentially small in n, by the Δ-continuation. Finally the error $O(\tau_0^2) = O(X/\mathcal{L})$ from Theorem 7.1 contributes a power-smaller coefficient error; even the weaker bound $O(X^{1-\epsilon})$, with $0 < \epsilon < 1/2$, suffices to put it beyond every displayed logarithmic order. No infinite asymptotic expansion has been interchanged with an integral.

For the first terms,

$$Q_1 = -A/2, \quad Q_2 = 3A^2/8 - A/2,$$

$$P_1 = -A/2 + p/2, \quad P_2 = 3A^2/8 - 3Ap/4 - A/2 + p/2 + 3g_2/8.$$

Writing $D = A - p$ gives $P_1 = -D/2$ and $P_2 = 3D^2/8 - D/2 - 3\psi^{(1)}(-1/2)/8$, proving (7.12). Formula (7.8) shows that the highest A coefficient is $\binom{-1/2}{m}$; all $j > 0$ terms in (7.9) have lower degree. Exact dominance follows as before from the nonzero singular term and Proposition 3.1. □

The two elementary checks $g_1 = 2 - \gamma - 2\log 2$ and $\psi^{(1)}(-1/2) = 4 + \pi^2/2$ follow from Gamma recurrence, duplication, and their derivatives [4]. They explain why Euler's constant and polygamma data appear even though the original equation contains only a polylogarithm and a quadratic polynomial.

8 Finite curvature does not always mean a Puiseux germ

For $s > 3$ at criticality put

$$C = 1 + \zeta(s-2) > 0, \qquad b = \frac{C}{2B} > 0. \tag{8.1}$$

Equation (4.2) has $H(\tau) = b\tau^2(1 + o(1))$. This gives the leading square root and (1.6). It does not imply that the whole inverse is an analytic function of $\sqrt{X}$.

8.1 Every noninteger order above three

Theorem 8.1 (A second convergent critical chart)**.** *Let $s > 3$ be noninteger and $u = U_{\mathrm{c}}(s)$. Put*

$$G = \Gamma(1-s), \qquad q = (X/b)^{1/2}, \qquad \sigma = q^{s-3}.$$

Then $\tau = qU(q,\sigma)$ for a unique analytic U near $(0,0)$ with $U(0,0)=1$. Its exact equation is

$$1 = U^2 + \frac{G}{Bb}\sigma U^{s-1} - \frac{q}{B}U + \frac{b}{2B}q^2 + \frac{1}{Bb}\sum_{k\geq 3}\frac{(-1)^k\zeta(s-k)}{k!}q^{k-2}U^k. \tag{8.2}$$

The support in X is contained in

$$\left\{\frac{1+j+k(s-3)}{2} : j,k\in\mathbb{N}_0\right\}. \tag{8.3}$$

Lemma 6.2, now with $E_0(U)=U^2-1$, gives every coefficient by a finite formula. The expansion converges, and its finite coefficient transfer uses Lemma 4.1 exactly as in Corollary 6.3.

Relative to the analytic square-root core obtained by deleting the Gamma term in (4.3), the first additional local term in v is

$$\frac{G}{2B}\,b^{-s/2}X^{(s-2)/2}. \tag{8.4}$$

If $s=a/d$ in lowest terms is noninteger, the minimum Puiseux ramification index is $2d$. If s is irrational, no finite ramification exists.

Proof. Substitute $X=bq^2$ and $\tau=qU$ into (4.5) with $w=B$, and divide by Bbq^2. Since $Bb=C/2$, this is (8.2). The equation is analytic in q,σ,U near $(0,0,1)$, and its U derivative at that point is 2. The analytic implicit function theorem gives the claimed convergent chart. Its two positive exponent weights give local finiteness of the support and finite transfer to any required asymptotic order.

The coefficient of σ in U at $q=0$ is $-G/(2Bb)$. Therefore the first term involving the new generator in $v=v_{\mathrm{b}}+X-qU$ is $Gq^{s-2}/(2Bb)$, which is (8.4). The remaining terms with at least one σ have strictly higher exponent; the $\sigma=0$ chart is precisely the analytic square-root reference core.

For rational $s=a/d$, the grid is contained in $(2d)^{-1}\mathbb{N}$ and gives an analytic series in $X^{1/(2d)}$. The nonzero leading square root and the nonzero exponent $(s-2)/2$ in (8.4) force the ramification index to be divisible by both 2 and the denominator of $(a-2d)/(2d)$. Their least common multiple is $2d$. The extra exponent cannot cancel with a pure half-integer term because s is noninteger. For irrational s, that exponent is irrational and nonzero, proving nonramifiability. □

The leading local term is $-b^{-1/2}X^{1/2}$. Its coefficient transfer is

$$\frac{b^{-1/2}}{2\sqrt{\pi}}R^{-n}n^{-3/2} = \sqrt{\frac{B}{2\pi(1+\zeta(s-2))}}R^{-n}n^{-3/2},$$

which proves (1.6) at every noninteger $s>3$. It also establishes exact dominance by the global proposition.

8.2 Integer orders retain a logarithmic obstruction

Theorem 8.2 (First resonant logarithm above the marginal order)**.** *Let $m\geq 4$ be an integer and $u=U_{\mathrm{c}}(m)$. Write*

$$B=\zeta(m-1),\quad b=\frac{1+\zeta(m-2)}{2B},\quad \ell_m=\frac{(-1)^m}{(m-1)!}.$$

There is a function $V(q)$ analytic near $q = 0$, with $V(\sqrt{X}) = v_{\mathrm{b}} - b^{-1/2}X^{1/2} + O(X)$, such that

$$v = V(\sqrt{X}) + \frac{\ell_m}{4B} b^{-m/2} X^{(m-2)/2} \log X + O\left(X^{(m-1)/2}(1 + |\log X|^2)\right). \tag{8.5}$$

The displayed logarithmic coefficient is nonzero. Consequently no finite Puiseux ramification represents the full inverse, despite its leading square root. The leading coefficient law is still (1.6)*.*

Proof. Write (4.4) as $\mathrm{Li}_m(\mathrm{e}^{-\tau}) = Z_m(\tau) + \ell_m \tau^{m-1} \log \tau$, with Z_m analytic. The corresponding analytic part of H has a double zero with leading coefficient b, so its inverse $\tau_a(X)$ is a convergent series in $\sqrt{X}$, beginning with $b^{-1/2}X^{1/2}$. Expanding the nonzero square root in (4.2) gives

$$H(\tau) - H_a(\tau) = \frac{\ell_m}{B} \tau^{m-1} \log \tau + O\left(\tau^m (1 + |\log \tau|^2)\right).$$

Inverting around τ_a, whose forward derivative is $2b\tau_a(1 + O(\tau_a))$, gives

$$\tau - \tau_a = -\frac{\ell_m}{2Bb} \tau_a^{m-2} \log \tau_a + O\left(\tau_a^{m-1}(1 + |\log \tau_a|^2)\right).$$

The possible second perturbation term has order $\tau_a^{2m-5} \log^2 \tau_a$, which is no larger than the stated remainder for $m \geq 4$. Now $\log \tau_a = \frac{1}{2} \log X + \log(\tau_a/\sqrt{X})$; the latter factor is analytic in $\sqrt{X}$. Absorb its leading analytic contribution into V and use $v = v_{\mathrm{b}} + X - \tau$. This yields (8.5).

The coefficient of the first logarithm cannot be removed by a function analytic in any finite root of X: on the common finite cover all such functions have ordinary power series, whereas the first nonzero logarithm still has nontrivial logarithmic continuation. Equivalently its asymptotic expansion has a logarithmic monomial absent from every Puiseux series. The leading square root, sectorial inversion, and the global proposition prove the coefficient law and unique dominance. □

Corollary 8.3 (A nonzero n^{-2} correction at order four)**.** *At $s = 4$, $u = U_{\mathrm{c}}(4)$, with $B = \zeta(3)$ and $C = 1 + \zeta(2)$,*

$$a_n R^n = \sqrt{\frac{B}{2\pi C}}\, n^{-3/2} + \frac{B}{6C^2}\, n^{-2} + O(n^{-5/2}(\log n)^2). \tag{8.6}$$

Proof. The logarithmic coefficient in (8.5) is $\ell_4 b^{-2}/(4B) = B/(6C^2)$. Its term is $X \log X$ and transfers with multiplier 1 by (4.7). The analytic X term does not contribute. The remaining Puiseux terms and logarithmic remainder have the stated coefficient order. □

Together the stable, marginal, and finite-curvature arguments finish the proof of Theorem 1.1. They also show why it is necessary to distinguish a leading square-root law from an analytic fold: at the polylogarithm boundary the forward function itself need not be analytic.

9 Finite-prefix changes and a normalization dictionary

The pure polylogarithm is not essential to the boundary proof. Finite changes in the low-index coefficients can be retained exactly without changing the tail class. This is useful when comparing separate packages, where a polynomial prefix or a different amplitude is often present.

Let

$$f_s(t) = \kappa \,\mathrm{Li}_s(t) + P(t), \qquad \kappa > 0, \quad P(0) = 0, \tag{9.1}$$

where P is a real polynomial. Assume that the final Taylor coefficients f_j are nonnegative and $f_1 > 0$. The coefficients of P need not all be nonnegative. Replace (1.1) by $uv - v^2/2 = f_s(re^v)$. Put $\mathcal{D} = t\,\mathrm{d}/\,\mathrm{d}t$ and define

$$\begin{aligned} A &= \kappa\zeta(s) + P(1), & B &= \kappa\zeta(s-1) + \mathcal{D}P(1), \\ U_c &= \sqrt{2A + B^2}, & w &= \sqrt{u^2 - 2A}, \quad v_b = u - w, \quad R = e^{-v_b}. \end{aligned} \tag{9.2}$$

Theorem 9.1 (Finite-prefix boundary transport)**.** *For $s > 2$, $u \geq U_c$, the exact radius is R, the only dominant singularity is R, and $\sum a_n R^n = v_b$. If $u > U_c$, then*

$$a_n R^n \sim \frac{\kappa}{w(1 - B/w)^s} n^{-s}. \tag{9.3}$$

At $u = U_c$, the fractional law for $2 < s < 3$ is

$$a_n R^n \sim \frac{(B/(\kappa\Gamma(1-s)))^{1/(s-1)}}{-\Gamma(-1/(s-1))} n^{-1-1/(s-1)}. \tag{9.4}$$

For $s > 3$, put $C = 1 + \kappa\zeta(s-2) + \mathcal{D}^2 P(1) > 0$. Then

$$a_n R^n \sim \sqrt{\frac{B}{2\pi C}}\, n^{-3/2}. \tag{9.5}$$

The convergent noninteger charts remain valid with $G = \kappa\Gamma(1-s)$ and with each regular coefficient $\zeta(s-k)$ replaced by $\kappa\zeta(s-k) + \mathcal{D}^k P(1)$.

At the marginal order $s = 3$, define

$$K = \frac{B}{\kappa}, \qquad b_0 = \frac{3}{2} + \frac{1 + \mathcal{D}^2 P(1)}{\kappa}. \tag{9.6}$$

The exact Lambert core is

$$Y = -W_{-1}(-4Ke^{-2b_0}X), \qquad \tau_0 = 2\sqrt{\frac{KX}{Y}}, \tag{9.7}$$

with the same continuation convention as in Theorem 7.1. The complete formula (7.10) holds with prefactor $\sqrt{K/\pi}$ and $A_n = \log\log n + 2b_0 - \log(4K)$. Its first two corrections are (7.12) with

$$D_n = \log\log n + 2b_0 - 2 + \gamma - \log K. \tag{9.8}$$

Proof. The lift is $h = u - \sqrt{u^2 - 2f_s(t)}$, with $uh_j = f_j + \frac{1}{2}\sum h_k h_{j-k}$. Nonnegative coefficients and $f_1 > 0$ imply $h_j \geq 0$, $h_1 > 0$, and $a_n > 0$ for every n by (2.3); the term using h_1^n already proves strictness. Also $A > 0$ and $B > 0$. The proof of Proposition 3.1 applies with Li_s replaced by f_s and Li_{s-1} by $\mathcal{D}f_s$. In particular $\mathcal{D}f_s(t) < B$ for $0 < t < 1$, and the strict circle argument is unchanged.

The regular Taylor coefficients of $P(e^{-\tau})$ are $(-1)^k \mathcal{D}^k P(1)/k!$, whereas the singular amplitude is multiplied by κ. This proves all noninteger statements by the same exact substitutions and implicit charts. At integer orders use the cancelled formula (4.4). When $s > 3$, $C = 1 + \sum j^2 f_j > 0$, so the curvature is nonzero.

At $s = 3$ the master equation has leading form

$$BX = \frac{\kappa\tau^2}{2}\left(\log\frac{1}{\tau} + \frac{3}{2} + \frac{1 + \mathcal{D}^2 P(1)}{\kappa}\right) + O(\tau^3 \log(1/\tau)).$$

This is $X = \tau^2(\log(1/\tau) + b_0)/(2K)$ plus the same power-smaller remainder. Solving it gives (9.7); the finite Lambert and Hankel calculations are universal once K and b_0 are substituted. This proves (9.8) and the all-order claim. □

For the pure model, $\kappa = 1$ and $P = 0$, so $K = B$ and $b_0 = 5/2$. For the numerical prefix test in the package, $f_s(t) = 2\,\mathrm{Li}_s(t) + t/2$, one has at $s = 3$

$$B = 2\zeta(2) + \tfrac{1}{2}, \qquad K = \zeta(2) + \tfrac{1}{4}, \qquad b_0 = \tfrac{9}{4}.$$

Thus neither the radius nor the logarithmic shift can be copied from the pure model without adjustment, even though the leading logarithmic exponent is unchanged.

10 Verification and reproducible evidence

The program `code/verify.py` separates exact finite tests from floating-point diagnostics. It writes only to an absent or empty output directory, so a replay does not overwrite the delivered evidence.

10.1 Exact checks

The recorded run passed **293 exact assertions**. For $s = 2, 3, 4$, $u = 1, 2, 3$, and $1 \le n \le 10$, rational arithmetic compares three independent coefficient constructions: iteration of the original quadratic fixed-point equation, the lift/Lagrange identity, and the predecessor's positive finite formula. Positivity is checked at each instance. Symbolic tests additionally compare the Lambert polynomials $Q_0, \dots, Q_4$ with a separate implicit recurrence, and the Hankel polynomials $P_0, \dots, P_4$ with direct substitution $L \mapsto L - y$ followed by moment replacement. The first two shifted corrections, the $s = 5/2$ local coefficient equations, the $s = 4$ logarithmic balance, and the leading polynomial coefficients are also checked.

These finite computations are regressions, not proofs of the universal claims. In particular, they do not replace the positive-circle or sectorial-continuation arguments.

10.2 Positive numerical coefficient recurrence

For each target index n define

$$p_k = [t^k] \exp(n(h(t) - h(1))).$$

Then

$$p_0 = \mathrm{e}^{-nv_{\mathrm{b}}}, \qquad p_k = \frac{n}{k} \sum_{j=1}^{k} j h_j p_{k-j}, \qquad a_n R^n = \frac{p_n}{n}. \tag{10.1}$$

The stored diagnostics use this positive recurrence, not the asymptotic formula, to compute the reference coefficients. Seven models were tested at seven indices up to 4096, for 49 coefficient evaluations. The platform's NumPy `longdouble` has 18 decimal digits of precision; these larger-index computations were *not* performed at 85 digits. For each model an independent arbitrary-precision recurrence at $n = 32$ agreed with the extended-precision result to relative error below $1.3 \cdot 10^{-18}$. The arbitrary-precision constants and local coordinate checks used `mpmath` at 85 decimal digits.

On platforms whose `longdouble` exponent range is too small for p_0, the code automatically uses an arbitrary-exponent `mpmath` recurrence. This fallback is slower but avoids silently underflowing the initial probability. Neither implementation supplies interval enclosures.

The marginal leading law approaches its limit slowly. Its corrections are therefore useful rather than decorative. Table 2 shows the coefficient divided by the refined approximation. In its $s = 3$ column both inverse-logarithmic corrections in (7.12) are retained; the fractional column retains the second nonzero term in (6.8). The reference coefficients in both columns still come from (10.1).

Table 1: Coefficient divided by its proved leading asymptotic. Strict boundary tests use $w = 2\zeta(s-1)$, hence $1 - B/w = 1/2$.

Regime	$n = 128$	$n = 512$	$n = 2048$	$n = 4096$
Critical $s = 5/2$	0.986604	0.994512	0.997779	0.998591
Critical $s = 3$	0.655458	0.703836	0.739120	0.753441
Critical $s = 4$	1.006135	1.003630	1.002010	1.001472
Strict, $s = 5/2$	0.906424	0.973409	0.993100	0.996528
Strict, $s = 3$	1.319718	1.092013	1.026750	1.014462
Strict, $s = 4$	1.425213	1.071592	1.016618	1.008208

Table 2: Independent coefficient divided by the refined approximation.

n	Critical $s = 5/2$, two nonzero terms	Critical $s = 3$, two corrections
128	1.001199056	1.049782812
512	1.000298579	1.020656682
2048	1.000074533	1.010075657
4096	1.000037253	1.007282765

There are 21 real or complex checks of the exact coordinate identity against the original equation, with recorded absolute residuals below 10^{-84}, and six checks of the defining Lambert-core identity. Local ratios at $\tau = 10^{-4}, 10^{-8}, 10^{-12}$ test the predicted leading normal forms; the data retain the observed errors rather than relabeling them as certified bounds. The prefix example $2\,\mathrm{Li}_3(t) + t/2$ tests the change of marginal normalization separately. Full values and software versions are in `data/verification.json` and the CSV files.

11 Source audit, corrections, and integration

11.1 What was actually inspected

The three user-specified areas were inspected through their GitHub indexes and selected source material [10, 9, 11]. The observed main commit was

`6207c3d40047fd8d1400a4d509fee71c9a3a8132`.

Initial index reads were on `main`; the later linear-core TeX read was pinned to this commit. Thus this is not falsely described as an atomic full-repository checkout. The incoming directory and commit metadata list five new polylogarithm/transseries archives. Their complete contents were not all audited.

The predecessor [1] was read in a Library TeX copy named `article(20261012-044824).tex`, specifically source lines 1–1000 and 1030–1487. Its exact title, equation, interior theorem, and pending boundary question identify the research target. The Library copy was not byte-compared with a member of `docs/incoming/Resonant_Polylogarithm_Transseries.zip`; that provenance limitation is explicit. The existing critical-Hahn package's README and its TeX opening through source line 150 were also inspected [8]. Its extensive editorial cross-references demonstrate that generic critical/Hahn and marginal mechanisms already belong to the repository's research program.

The external source check covered the DLMF polylogarithm formula and Lambert material, the Janson publication record/abstract, and targeted literature searches. Direct retrieval of the full Flajolet–Odlyzko article was unsuccessful; it is cited as the classical transfer source, not represented as newly audited line by line. The needed local transfer and finite Hankel

calculations are explained in this manuscript. Neither the repository inspection nor the literature search establishes worldwide publication priority.

11.2 Status changes and safeguards, not invented source errors

No new mathematical error was established in the accepted repository statements inspected. In particular, the predecessor correctly labels (1.3) as unproved and correctly restricts its existing fold theorem to the interior regime. The appropriate proposed editorial change is to mark its boundary question as answered *for fixed parameters* by this article, subject to independent proof review. Its joint-limit questions remain open.

There are four useful integration safeguards. First, the simple-boundary formula cannot be continued by substitution to criticality: its constant diverges. Second, inserting $t = 1$ into the interior curvature prefactor at $s = 3$ would incorrectly erase the leading coefficient; the missing factor is $(\log n)^{-1/2}$. Third, above $s = 3$ a leading square root is not a theorem of analytic Puiseux structure: Theorem 8.2 supplies an explicit nonzero logarithmic obstruction at every integer $m \geq 4$. Fourth, neglecting Gamma zeros creates a false n^{-2} contribution at $s = 5/2$, whereas at $s = 4$ the logarithmic term really does produce such a contribution. These are precise consequences of the present proofs, not accusations about unseen source passages.

11.3 Suggested placement and statement-level review

A self-contained placement is

`Analysis/Transseries/docs/series-and-transseries/`
`Quadratic_Polylogarithm_Boundary_Transfer/`.

The polylogarithm project index can link to this package, and the pending article can receive a short editorial cross-reference rather than having its original proofs silently rewritten. `INTEGRATION.md` and `integration_note.tex` supply proposed wording. No remote change was made.

A sensible review order is the exact coefficient identity and the master equation, then the positive-circle argument, then the strict-boundary proof and integer cancellation. The two convergent critical charts and the Lambert/Hankel compiler can be reviewed independently afterward. The proposed results do not establish general resurgence, Borel summability, a uniform moving-critical-point law, or any new Lean theorem.

12 Further research questions

1. Uniform approach to the critical surface. Theorem 5.1 and the critical laws are fixed-parameter statements. For $2 < s < 3$, balancing the drift term $(w - B)\tau$ against $\Gamma(1 - s)\tau^{s-1}$ in the master equation suggests a coefficient crossover when $w - B$ is of order $n^{-(s-2)/(s-1)}$. Prove a uniform coefficient profile, including its normalizing constant and both outer limits. The balance is a scaling suggestion, not a proved uniform theorem.

2. The joint order-three limit. Construct a pole-cancelled inverse chart for $s \to 3$ and $n \to \infty$ that matches the two Hahn charts with the Lambert chart. The Gamma pole and the pole of $\zeta(s - 2)$ must remain paired. The linear-core marginal and confluent packages are comparison sources; the quadratic drift terms and normalization b_0 must be tracked explicitly.

3. The lower endpoint and large-core limits. As $s \downarrow 2$, the moment $B = \zeta(s-1)$ diverges and the critical surface escapes to infinity. Determine which joint scalings connect this boundary theory to the dilogarithm's interior fold and to the predecessor's large-core, large-sector question. Neither the fixed-s formulas nor a formal limit of their constants provides that connection.

4. Parameter derivatives and resonance cancellation. Differentiate the exact coefficient formulas with respect to s and u. Can one construct a uniform order-jet calculus for coefficients near rational Hahn collisions and integer logarithmic resonance? The transfer dictionary suggests explicit polygamma and $\log n$ terms, but uniform control must precede differentiating an asymptotic error term.

5. Other exact cores. Replace $uv - v^2/2$ by the higher-degree exact polynomial cores of integer-order Fermi inversion. The quadratic lift no longer applies directly, and positivity may fail. Identify conditions on a general analytic core that preserve the unique positive dominant boundary, and determine how its first nonzero derivatives enter the marginal shift.

6. Periodic or signed tails. Replace the positive tail by a periodic-weight or cyclotomic polylogarithm combination. Several boundary points can compete and the strict triangle-inequality proof may break down. A finite Fourier criterion for dominant-point multiplicity and cancellation would connect this theory to the predecessor's periodic Borel-residue dictionary.

7. Effective convergence domains and error certificates. The analytic implicit-function proofs give genuine local convergence but not optimal explicit polydisks. Produce interval-certified domains for the two-variable charts and for the Lambert perturbation, and certified coefficient enclosures for the positive recurrence. The floating-point diagnostics delivered here are not such certificates.

8. Formalization and exact coefficient infrastructure. The finite reversion formula and the rational coefficient recurrence are natural first proof-assistant targets. A second layer would formalize the polylogarithmic transfer dictionary and its integer limit. The full radius theorem additionally requires positive implicit-series convergence, complex continuation, and Hankel transfer. The theorem manifest separates these dependencies so that finite algebraic checks are not mistaken for formal analytic coverage.

Conclusion

The pending quadratic boundary law is an analytic theorem once positivity, sectorial inversion, and integer-order pole cancellation are retained together. The critical boundary carries more information than a leading exponent: it has convergent Hahn charts away from integer resonance, Gamma-controlled missing coefficient terms, and an exactly normalized Lambert core at the marginal order. A finite quadratic term changes the logarithmic constants and can produce resonant corrections invisible in a naive square-root description. These refinements connect the exact polylogarithm calculus to the repository's transseries inversion program without claiming its classical or already-developed mechanisms as new.

A Reproduction and artifact map

The article is self-contained in `article.tex`; compilation does not require external images or font files. A TeX Live installation with the standard packages named in the preamble is sufficient. The retained numerical tables are rounded from the accompanying CSVs.

```
python -m pip install -r requirements.txt
python code/verify.py --output-dir replay --max-n 4096
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The output directory must be new or empty. The distributed `data/verification.json` describes the executed run, not an unperformed command. `BUILD_VALIDATION.json` records the PDF build and page-layout checks. `THEOREMS.json` lists theorem dependencies and the distinction between conventional proofs and finite checks.

Artifact	Purpose
`article.tex`, `article.pdf`	Complete proofs, exact identities, research questions, and source distinctions.
`code/verify.py`	Exact finite comparisons and positive numerical coefficient diagnostics.
`data/exact_coefficients.csv`	Rational coefficients for the finite replay cases.
`data/marginal_polynomials.txt`	$Q_0, \dots, Q_4$ and $P_0, \dots, P_4$ in symbolic Gamma-moment coordinates.
`data/coefficient_diagnostics.csv`	49 positive-recurrence comparisons through $n = 4096$, with arithmetic engine recorded.
`data/local_checks.csv`, `data/lambert_checks.csv`	Local normal-form and defining-equation diagnostics.
`SOURCES.md`, `INTEGRATION.md`	Provenance, scope limits, and proposed filing/editorial notes.

References

[1] Research draft prepared for Vladimir Reshetnikov with OpenAI assistance, *Resonant Polylogarithm Transseries: Gamma Borel Kernels, Cyclotomic Order Jets, and Sharp Dilogarithm Reversion*, 11 October 2026. Theorem 9.1 and Section 12, Question 1. Inspected Library TeX copy `article(2026 1012-044824).tex`; associated incoming archive name `docs/incoming/Resonant_Polylogarithm _Transseries.zip`. Source ranges and provenance limitation are stated in Section 11.

[2] NIST Digital Library of Mathematical Functions, *§25.12, Polylogarithms*, especially 25.12.10–25.12.12. https://dlmf.nist.gov/25.12. Inspected 12 October 2026. The Gamma–zeta expansion is attributed there to Erdélyi et al., *Higher Transcendental Functions*, vol. 1 (1953), Section 1.11.

[3] NIST Digital Library of Mathematical Functions, *§4.13, Lambert W-Function*, branch definitions and asymptotic expansions. https://dlmf.nist.gov/4.13. Inspected 12 October 2026.

[4] NIST Digital Library of Mathematical Functions, *Chapter 5, Gamma Function*, especially recurrence, duplication, and polygamma relations in §§5.5 and 5.15. https://dlmf.nist.gov/5. Classical identities used in (7.7); the finite recurrence evaluations are reproduced in `code/verify.py`.

[5] P. Flajolet and A. Odlyzko, Singularity analysis of generating functions, *SIAM Journal on Discrete Mathematics* **3** (1990), 216–240. https://doi.org/10.1137/0403019. Classical transfer reference; direct full-text retrieval was unsuccessful in the present audit.

[6] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapters VI and VII. Standard reference for singularity transfer and positive implicit-function schemas; not represented as a new full-book audit.

[7] S. Janson, Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation, *Probability Surveys* **9** (2012), 103–252. https://arxiv.org/abs/1112.0510. Publication record and abstract inspected; used to identify the established probabilistic framework, not as a substitute for the proofs above.

[8] ProveIt research package, *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, 29 September 2026, with editorial amendments. `Anal ysis/Transseries/docs/series-and-transseries/ Critical_Hahn_Transseries_Beyond_F inite_Action_Folds/`. README and TeX source lines 1–150 inspected; the latter at commit `6207c3d40047fd8d1400a4d509fee71c9a3a8132`.

[9] ProveIt Contributors, *Series and transseries: canonical-volume and research-package index*. `Anal ysis/Transseries/docs/series-and-transseries/README.md`, lines 1–220 inspected on 12 October 2026.

[10] ProveIt Contributors, *Polylogarithms project index*. `Analysis/Polylogarithms/README.md`, lines 1–190 inspected on 12 October 2026. Accepted/pending status is governed by the project ledger, not by a historical article abstract alone.

[11] ProveIt Contributors, *Incoming research intake instructions and archive index*. `docs/incoming/ README.md`, lines 1–145, and directory/commit metadata inspected on 12 October 2026. https: //github.com/VladimirReshetnikov/ProveIt.